

SHRUTI JAIN

+1-412-589-5683 | spjain@andrew.cmu.edu | [in shruti-jain-1405](https://www.linkedin.com/in/shruti-jain-1405) | shrutijain1405.github.io

EDUCATION

Carnegie Mellon University | Master of Science in Computer Vision Aug 2025 - Expected Dec 2026
Coursework: Learning for 3D, Geometry based Vision, Advanced Computer Vision, Introduction to Machine Learning
CGPA: **4.17/4**

Manipal Institute of Technology | Bachelor of Technology in Information Technology 2019 - 2023
CGPA: **3.48/4**

SKILLS

Programming Languages/Tools: Python, C++, Java, Docker, Kubernetes, SQL, Temporal
ML and CV Frameworks/Libraries: PyTorch, OpenCV, ROS, CUDA, scikit-learn, TensorFlow, Matplotlib
AI Topics: Multimodal Learning, Vision-Language Models, AI Agents, Diffusion Models, Visual Grounding, Post-Training (LoRA, GRPO), Object Tracking, Image Segmentation, Prompt Optimization, 3D Gaussian Splatting

RELEVANT EXPERIENCE

AIRLab - Carnegie Mellon University | *Research Assistant* Jan 2026 - Present

- [\[Project Website\]](#) Developed a multimodal medical triage pipeline for the [DARPA Triage Challenge](#) using RGB and thermal UAV imagery to predict **50+** hierarchical medical labels, under the guidance of **Prof. Sebastian Scherer**.
- Built a two-stage system combining conditional **diffusion-based RGB-to-IR generation** and VLM fine-tuning with **LoRA/GRPO optimization** to improve triage reasoning under limited paired RGB-IR data.
- Designing **post-training strategies** for multimodal RGB-IR reasoning with **Cosmos Reason 2** and building a **Nemotron 3 based orchestrator** to coordinate specialized models across the medical-triage pipeline.

FlowState AI | *Computer Vision Research Intern* May 2026 - Aug 2026

- [\[Slides\]](#) **Architected and shipped** an end-to-end **AI video-editing system** from concept to production, transforming long-form footage into prompt-driven, story-based vertical content with intelligent 16:9-to-9:16 reframing.
- Built a **self-correcting video pipeline** combining **VLM** subject identification and grounding, **SAM3** tracking, and trajectory verification, with scene and shot-level orchestration in **Temporal**.
- Isolated and evaluated individual pipeline components through **targeted benchmarks and ablation studies**, achieving **80.9% precision** at a 20-pixel threshold for ball tracking on **SoccerNet**.

Carnegie Mellon University, USA | *Research Assistant* Aug 2025 - Dec 2025

- Researched under **Prof. Deva Ramanan** and developed DetPO, a gradient-free black-box prompt optimization framework achieving up to **9.7% improvement** over prior approaches on Roboflow20-VL and LVIS.

Palo Alto Networks | *Software Developer* Oct 2023 - Apr 2025

- Worked on features and APIs of the microservices (**Java Spring Boot**) for ingesting events from tenants' cloud accounts (**AWS, Azure, GCP**) for near real-time threat detection using **anomaly detection**.

Robotics Research Centre (RRC) - IIIT Hyderabad | *Research Assistant* Jun 2023 - Sept 2023

- Achieved **14.4%** average decrease in the mean absolute scores across 3 datasets as compared to the SOTA for **Image Segmentation in self-driving cars** in adverse weather conditions.
- Implemented a **differentiable visual and latent prompting mechanism** to fine-tune on **Segment Anything Model** via **adaptors** through learnable image pre-processing blocks.

PUBLICATIONS

- DetPO: In-Context Learning with Multi-Modal LLMs for Few-Shot Object Detection [\[Paper\]](#) [\[Website\]](#) [\[Code\]](#)
Published in ECCV 2026
- DiffPrompter: Differentiable Implicit Visual Prompts for Semantic-Segmentation in Adverse Conditions [\[Paper\]](#) [\[Project Page\]](#) [\[Code\]](#) *Published in IROS 2024*
- Light Weight Character and Shape Recognition for Autonomous Drones [\[Paper\]](#)
Accepted at Workshop, ICML 2022
- A Studious Approach To Semi-Supervised Learning [\[Paper\]](#) *Accepted at ICBINB Workshop, NeurIPS 2021*

RELEVANT PROJECTS

- **Gated Multimodal Web Navigation Agents** - [\[Poster\]](#) A gated multimodal web navigation framework combining MindAct and Qwen2-VL on the Multimodal Mind2Web benchmark, improving robustness to adversarial popup attacks.
- **Editing 3D Gaussian Splats** - [\[Slides\]](#) Built a **text-guided 3D Gaussian Splat editing** pipeline integrating **DDIM inversion** and **InstructPix2Pix** with **cross-view attention** to preserve multi-view consistency, supporting global scene edits and localized object edits through language-guided segmentation masks.
- **FlashAttention for ViT** - [\[Poster\]](#) Implemented FlashAttention-1 and 2 in MiniTorch as custom CUDA autograd ops with tiled SRAM compute and online softmax.
- **Real-time Failed 3D Print Detector** - Detects failed 3D prints in real time using OpenCV and **anomaly detection**.